

# 1 NAME OF THE MEDICINAL PRODUCT

ACARIZAX® 12 SQ-HDM sublingual lyophilisate

## 2 QUALITATIVE AND QUANTITATIVE COMPOSITION

Standardised allergen extract from the house dust mites 12 SQ-HDM\* per sublingual lyophilisate

6 SQ-HDM *Dermatophagoides pteronyssinus*

6 SQ-HDM *Dermatophagoides farinae*

For a full list of excipients, see section 6.1

\* [SQ-HDM is the dose unit for ACARIZAX®. SQ is a method for standardisation on biological potency, major allergen content and complexity of the allergen extract. HDM is an abbreviation for house dust mite.]

## 3 PHARMACEUTICAL FORM

Sublingual lyophilisate

White to off-white freeze-dried debossed sublingual lyophilisate

## 4 CLINICAL PARTICULARS

### 4.1 Therapeutic indications

ACARIZAX® is indicated in adult patients (18-65 years) diagnosed by clinical history and a positive test of house dust mite sensitisation (skin prick test and/or specific IgE) with at least one of the following conditions:

- persistent moderate to severe house dust mite allergic rhinitis despite use of symptom-relieving medication
- house dust mite allergic asthma not well controlled by inhaled corticosteroids and associated with mild to severe house dust mite allergic rhinitis. Patients' asthma status should be carefully evaluated before the initiation of treatment (see section 4.3).

ACARIZAX® is indicated in adolescents (12-17 years) diagnosed by clinical history and a positive test of house dust mite sensitisation (skin prick test and/or specific IgE) with persistent moderate to severe house dust mite allergic rhinitis despite use of symptom-relieving medication.

### 4.2 Posology and method of administration

#### Posology

The recommended dose for adults and adolescents (12-17 years) is one sublingual lyophilisate (12 SQ-HDM) daily.

Onset of the clinical effect is to be expected 8-14 weeks after initiation. International treatment guidelines refer to a treatment period of 3 years for allergy immunotherapy to achieve disease modification. Efficacy data is available for 18 months of treatment with ACARIZAX® in adults; no data is available for 3 years of treatment (see section 5.1). If no improvement is observed during the first year of treatment with ACARIZAX® there is no indication for continuing treatment.

#### Paediatric population

Clinical experience on immunotherapy with ACARIZAX® in children <12 years of age has not been established. ACARIZAX® is not intended for use in children <12 years of age. Currently available data in children are described in section 5.1.

#### Elderly population

Clinical experience on immunotherapy with ACARIZAX<sup>®</sup> in adults >65 years of age has not been established. ACARIZAX<sup>®</sup> is not intended for use in adults >65 years of age (see section 5.1).

#### Method of administration

ACARIZAX<sup>®</sup> treatment should be initiated by physicians with experience in treatment of allergic diseases. It is recommended that the first sublingual lyophilisate is taken under medical supervision and that the patient is monitored for at least half an hour, to enable discussion and possible treatment of any immediate side effects.

ACARIZAX<sup>®</sup> is an sublingual lyophilisate. The sublingual lyophilisate should be taken with dry fingers from the blister unit immediately after opening the blister and placed under the tongue, where it will disperse. Swallowing should be avoided for approximately 1 minute. Food and beverage should not be taken for the following 5 minutes.

If treatment with ACARIZAX<sup>®</sup> is interrupted for a period up to 7 days, treatment can be resumed by the patient. If the treatment is interrupted for more than 7 days it is recommended to contact a physician before resuming the treatment.

### **4.3 Contraindications**

Hypersensitivity to any of the excipients (for a full list of excipients, see section 6.1).

Patients with FEV<sub>1</sub> < 70% of predicted value (after adequate pharmacological treatment) at initiation of treatment.

Patients who have experienced a severe asthma exacerbation within the last 3 months.

In patients with asthma and experiencing an acute respiratory tract infection, initiation of ACARIZAX<sup>®</sup> treatment should be postponed until the infection has resolved.

Patients with active or poorly controlled autoimmune diseases, immune defects, immunodeficiencies, immunosuppression or malignant neoplastic diseases with current disease relevance.

Patients with acute severe oral inflammation or oral wounds (see section 4.4).

### **4.4 Special warnings and precautions for use**

#### Asthma

Asthma is a known risk factor for severe systemic allergic reactions.

Patients should be advised that ACARIZAX<sup>®</sup> is not intended to treat acute asthma exacerbations. In the event of an acute asthma exacerbation, a short-acting bronchodilator should be used. If patients find short-acting bronchodilator treatment ineffective or they need more inhalations than usual, medical attention must be sought.

Patients must be informed of the need to seek medical attention immediately if their asthma deteriorates suddenly.

ACARIZAX<sup>®</sup> should initially be used as add on therapy and not as a substitute of pre-existing asthma medication. Abrupt discontinuation of asthma controller medication after initiation of ACARIZAX<sup>®</sup>

treatment is not recommended. Reductions in asthma controller medication should be performed gradually under the supervision of a physician according to asthma management guidelines.

#### Severe systemic allergic reactions

Treatment should be discontinued and a physician should be contacted immediately in case of severe systemic allergic reactions, severe asthma exacerbation, angioedema, difficulty in swallowing, difficulty in breathing, changes in voice, hypotension or feeling of fullness in the throat. The onset of systemic symptoms may include flushing, pruritus, sense of heat, general discomfort and agitation/anxiety.

One option for treating severe systemic allergic reactions is adrenaline. The effects of adrenaline may be potentiated in patients treated with tricyclic antidepressants, mono amino oxidase inhibitors (MAOIs) and/or COMT inhibitors with possible fatal consequences. The effects of adrenaline may be reduced in patients treated with beta-blockers.

Patients with cardiac disease may be at increased risk in case of systemic allergic reactions. Clinical experience in treatment with ACARIZAX® of patients with cardiac disease is limited.

This should be taken into consideration prior to initiating allergy immunotherapy.

Initiation of ACARIZAX® in patients who have previously had a systemic allergic reaction to subcutaneous house dust mite immunotherapy should be carefully considered, and measures to treat potential reactions should be available. This is based on post-marketing experience from a corresponding sublingual tablet product for grass pollen immunotherapy which indicates that the risk of a severe allergic reaction may be increased for patients who have previously experienced a systemic allergic reaction to subcutaneous grass pollen immunotherapy.

#### Oral inflammation

In patients with severe oral inflammation (e.g. oral lichen planus, mouth ulcers or thrush), oral wounds or following oral surgery, including dental extraction, or following tooth loss, initiation of ACARIZAX® treatment should be postponed and ongoing treatment should be temporarily interrupted to allow healing of the oral cavity.

#### Local Allergic Reactions

When treated with ACARIZAX® the patient is exposed to the allergen that causes the allergic symptoms. Therefore, local allergic reactions are to be expected during the treatment period. These reactions are usually mild or moderate; however, more severe oropharyngeal reactions may occur. If the patient experiences significant local adverse reactions from the treatment, anti-allergic medication (e.g. antihistamines) should be considered.

#### Eosinophilic oesophagitis

Cases of eosinophilic oesophagitis have been reported in association with ACARIZAX® treatment. In patients with severe or persisting gastro-oesophageal symptoms such as dysphagia or dyspepsia, ACARIZAX® should be interrupted and medical evaluation must be sought.

#### Autoimmune diseases in remission

Limited data is available on treatment with allergy immunotherapy in patients with autoimmune diseases in remission. ACARIZAX® should therefore be prescribed with caution in these patients.

Food allergy

ACARIZAX<sup>®</sup> may contain trace amounts of fish protein. Available data have not indicated an increased risk of allergic reactions in patients with fish allergy.

#### **4.5 Interaction with other medicinal products and other forms of interaction**

No interaction trials have been conducted in humans and no potential drug interactions have been identified from any source. Concomitant therapy with symptomatic anti-allergic medications may increase the tolerance level of the patient to immunotherapy. This should be considered at discontinuation of such medications.

#### **4.6 Fertility, pregnancy and lactation**

##### Pregnancy

There is no data on the clinical experience for the use of ACARIZAX<sup>®</sup> in pregnant women. Animal studies do not indicate increased risk to the foetus. Treatment with ACARIZAX<sup>®</sup> should not be initiated during pregnancy. If pregnancy occurs during treatment, the treatment may continue after evaluation of the general condition (including lung function) of the patient and reactions to previous administration of ACARIZAX<sup>®</sup>. In patients with pre-existing asthma close supervision during pregnancy is recommended.

##### Lactation

No clinical data are available for the use of ACARIZAX<sup>®</sup> during lactation. No effects on the breastfed infants are anticipated.

##### Fertility

There is no clinical data with respect to fertility for the use of ACARIZAX<sup>®</sup>. In a repeat dose toxicity study in mice no effects were observed in the reproductive organs of both genders.

#### **4.7 Effects on ability to drive and use machines**

Treatment with ACARIZAX<sup>®</sup> has no or negligible influence on the ability to drive or use machines.

#### **4.8 Undesirable effects**

##### Summary of the safety profile

Subjects taking ACARIZAX<sup>®</sup> should primarily expect mild to moderate local allergic reactions to occur within the first few days and subsiding again with continued treatment (1-3 months) (see section 4.4). For the majority of events, the reaction should be expected to start within 5 minutes after intake of ACARIZAX<sup>®</sup> on each day of occurrence and abate after minutes to hours. More severe oropharyngeal allergic reactions may occur (see section 4.4).

*Tabulated list of adverse reactions*

The following table of adverse reactions is based on data from placebo-controlled clinical trials investigating ACARIZAX<sup>®</sup> in adult and adolescent patients with house dust mite allergic rhinitis and/or allergic asthma and/or from spontaneous reporting.

Adverse reactions are divided into groups according to the MedDRA convention frequencies: Very common ( $\geq 1/10$ ), common ( $\geq 1/100$  to  $< 1/10$ ), uncommon ( $\geq 1/1,000$  to  $< 1/100$ ), rare ( $\geq 1/10,000$  to  $< 1/1,000$ ), very rare ( $< 1/10,000$ ).

| System Organ Class                                   | Frequency   | Adverse Drug Reaction                                                                                                                                                                                                                                               |
|------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Immune system disorders                              | Uncommon    | Anaphylactic reaction                                                                                                                                                                                                                                               |
| Nervous system disorders                             | Common      | Dysgeusia                                                                                                                                                                                                                                                           |
|                                                      | Uncommon    | Dizziness                                                                                                                                                                                                                                                           |
| Eye Disorders                                        | Common      | Eye pruritus                                                                                                                                                                                                                                                        |
| Ear and labyrinth disorders                          | Very common | Ear pruritus                                                                                                                                                                                                                                                        |
|                                                      | Uncommon    | Ear discomfort                                                                                                                                                                                                                                                      |
| Respiratory, thoracic and mediastinal disorders      | Very common | Throat irritation                                                                                                                                                                                                                                                   |
|                                                      | Common      | Asthma*, cough <sup>#</sup> , dysphonia, dyspnoea, oropharyngeal pain, pharyngeal oedema                                                                                                                                                                            |
|                                                      | Uncommon    | Nasal congestion, rhinorrhoea, sneezing, throat tightness                                                                                                                                                                                                           |
|                                                      | Rare        | Laryngeal oedema                                                                                                                                                                                                                                                    |
| Gastrointestinal disorders                           | Very common | Lip oedema, oedema mouth, oral pruritus                                                                                                                                                                                                                             |
|                                                      | Common      | Abdominal pain, diarrhoea, dyspepsia, dysphagia, gastrooesophageal reflux disease, glossitis, glossodynia, lip pruritus, mouth ulceration, tongue pruritus, nausea, oral discomfort, oral mucosal erythema, paraesthesia oral, stomatitis, tongue oedema, vomiting. |
|                                                      | Uncommon    | Oesophageal irritation, oral mucosal blistering                                                                                                                                                                                                                     |
|                                                      | Rare        | Eosinophilic oesophagitis                                                                                                                                                                                                                                           |
| Skin and subcutaneous tissue disorders               | Common      | Pruritus, urticaria                                                                                                                                                                                                                                                 |
|                                                      | Uncommon    | Erythema, rash                                                                                                                                                                                                                                                      |
|                                                      | Rare        | Angioedema                                                                                                                                                                                                                                                          |
| General disorders and administration site conditions | Common      | Chest discomfort, fatigue                                                                                                                                                                                                                                           |
|                                                      | Uncommon    | Sensation of foreign body                                                                                                                                                                                                                                           |

#### *Description of selected adverse reactions*

If the patient experiences significant adverse reactions from the treatment, anti-allergic medication should be considered.

Cases of serious systemic allergic reactions, including anaphylaxis have been reported post marketing. The medical supervision at first sublingual lyophilisate intake is therefore an important precaution (see section 4.2). In some cases the serious systemic allergic reaction has occurred at doses subsequent to the initial dose.

In case of acute worsening in asthma symptoms or severe systemic allergic reactions, angioedema, difficulty in swallowing, difficulty in breathing, changes in voice, hypotension or feeling of fullness in the throat a physician should be contacted immediately. In such cases treatment should be discontinued permanently or until otherwise advised by the physician.

\*Isolated cases of severe acute worsening of asthma symptoms have been reported. Patients with known risk factors should not initiate treatment with ACARIZAX® (See section 4.3).

\*In clinical trials cough was observed with the same frequency for ACARIZAX® and placebo.

#### *Paediatric population*

ACARIZAX® is not indicated in patients <12 years of age. Only limited data from patients 5-11 years of age are available and no data on treatment with ACARIZAX® in children <5 years of age exist.

### **4.9 Overdose**

In phase I studies adult patients with house dust mite allergy were exposed to doses up to 32 SQ-HDM. No data is available in adolescents regarding exposure to doses above the recommended daily dose of 12 SQ-HDM.

If doses higher than the recommended daily dose are taken, the risk of side effects increases, including the risk of systemic allergic reactions or severe local allergic reactions. In case of severe reactions such as angioedema, difficulty in swallowing, difficulty in breathing, changes in voice, or feeling of fullness in the throat, immediate medical evaluation is needed. These reactions should be treated with relevant symptomatic medication.

## **5 PHARMACOLOGICAL PROPERTIES**

### **5.1 Pharmacodynamic properties**

Pharmacotherapeutic group: Allergen extracts, house dust mite

ATC code: V01AA03

### Mechanism of action

ACARIZAX<sup>®</sup> is allergy immunotherapy. Allergy immunotherapy with allergen products is the repeated administration of allergens to allergic individuals with the purpose of modifying the immunological response to the allergen.

The immune system is the target for the pharmacodynamic effect of allergy immunotherapy, but the complete and exact mechanism of action regarding the clinical effect is not fully understood. Treatment with ACARIZAX<sup>®</sup> has been demonstrated to induce an increase in house dust mite specific IgG4 and to induce a systemic antibody response that can compete with IgE in the binding of house dust mite allergens. This effect is observed already after 4 weeks of treatment.

ACARIZAX<sup>®</sup> works by addressing the cause of house dust mite respiratory allergic disease, and clinical effect during treatment has been demonstrated for both upper and lower airways. The underlying protection provided by ACARIZAX<sup>®</sup> leads to improvement in disease control and improved quality of life demonstrated through symptom relief, reduced need for other medications and a reduced risk for exacerbation.

### Clinical efficacy in adults

The efficacy of treatment with ACARIZAX<sup>®</sup> 12 SQ-HDM in house dust mite respiratory allergic disease was investigated in two double-blind, randomised, placebo-controlled trials with different endpoints and in different patient populations. Two thirds of the trial subjects were sensitised to more allergens than just house dust mite. Being sensitised to house dust mite only or to house dust mite and one or more other allergens did not impact the trial results. Supportive evidence from an allergen exposure chamber trial as well as a trial conducted with lower doses is also presented.

### ***Allergic rhinitis***

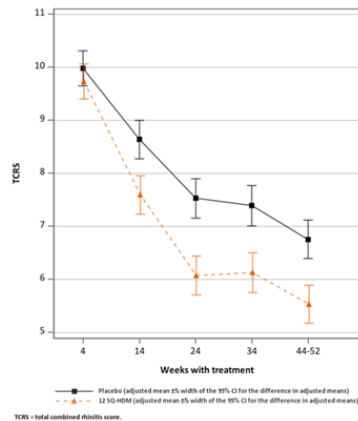
#### The MERIT trial (MT-06)

- The MERIT trial included 992 adults with moderate-to-severe house dust mite allergic rhinitis despite the use of rhinitis pharmacotherapy. Subjects were randomised to approximately 1 year of daily treatment with 12 SQ-HDM, 6 SQ-HDM or placebo and were given free access to standardised rhinitis pharmacotherapy. Subjects were seen by a specialist approximately every two months during the entire trial.
- The primary endpoint was the average daily total combined rhinitis score (TCRS) evaluated during the last 8 weeks of treatment.
  - The TCRS was the sum of the rhinitis symptoms score and the rhinitis medication score. The rhinitis symptoms score evaluated 4 nasal symptoms (runny nose, blocked nose, itching nose, sneezing) daily on a 0-3 scale (no, mild, moderate, severe symptoms), i.e. range of scale is 0-



12. The rhinitis medication score was the sum of the score for nasal steroid intake (2 points per puff, max. 4 puffs/day) and oral antihistamine intake (4 points/tablet, max. 1 tablet/day), i.e. range: 0-12. Thus the TCRS range is: 0-24.

- Additional pre-defined key secondary endpoints were the total combined rhinoconjunctivitis score and rhinoconjunctivitis quality of life (RQLQ).
- Post-hoc analyses of the days with a rhinitis exacerbation were also conducted to further illustrate the clinical relevance of the results.
  - A rhinitis exacerbation was defined as a day where the subject returned to the high level of symptoms required for trial inclusion: a rhinitis symptom score of at least 6 or at least 5 with one symptom rated severe.



**The MERIT trial: Development over time of the total combined rhinitis score**

TCRS: total combined rhinitis score (symptoms + medication score).

The primary endpoint was the average daily TCRS during the last approximately 8 weeks of treatment (weeks ~44-52).

Adjusted means of the average TCRS over time with error bars for the difference in adjusted means. Non-overlapping intervals indicate a statistically significant difference.

| MERIT results                                                                                           | 12 SQ-HDM |            | Placebo |            | Treatment effect                    |                                  |              |
|---------------------------------------------------------------------------------------------------------|-----------|------------|---------|------------|-------------------------------------|----------------------------------|--------------|
| Primary endpoint                                                                                        | N         | Score      | N       | Score      | Absolute difference <sup>c</sup>    | Relative difference <sup>d</sup> | p- value     |
| <b>Total combined rhinitis score</b>                                                                    |           |            |         |            |                                     |                                  |              |
| FAS-MI <sup>a</sup><br>(adjusted mean)                                                                  | 318       | 5.71       | 338     | 6.81       | 1.09<br>[0.35;1.84]                 | -                                | <b>0.004</b> |
| FAS <sup>b</sup><br>(adjusted mean)                                                                     | 284       | 5.53       | 298     | 6.76       | 1.22<br>[0.49;1.96]                 | 18%                              | <b>0.001</b> |
| FAS <sup>b</sup><br>(median)                                                                            | 284       | 5.88       | 298     | 7.54       | 1.66                                | 22%                              | -            |
| Pre-defined key secondary endpoints                                                                     | N         | Score      | N       | Score      | Absolute difference <sup>c</sup>    | Relative difference <sup>d</sup> | p- value     |
| <b>Rhinitis symptoms score</b>                                                                          |           |            |         |            |                                     |                                  |              |
| FAS <sup>b</sup><br>(adjusted mean)                                                                     | 284       | 2.76       | 298     | 3.30       | 0.54<br>[0.18;0.89]                 | 16%                              | <b>0.003</b> |
| FAS <sup>b</sup><br>(median)                                                                            | 284       | 2.98       | 298     | 3.98       | 1.00                                | 25%                              | -            |
| <b>Rhinitis medication score</b>                                                                        |           |            |         |            |                                     |                                  |              |
| FAS <sup>b</sup><br>(adjusted mean)                                                                     | 284       | 2.22       | 298     | 2.83       | 0.60<br>[0.08;1.13]                 | 21%                              | <b>0.024</b> |
| FAS <sup>b</sup><br>(median)                                                                            | 284       | 2.83       | 298     | 4.00       | 1.17                                | 29%                              | -            |
| <b>Total combined rhinoconjunctivitis score</b>                                                         |           |            |         |            |                                     |                                  |              |
| FAS <sup>b</sup><br>(adjusted mean)                                                                     | 241       | 7.91       | 257     | 9.12       | 1.21<br>[0.13;2.28]                 | 13%                              | <b>0.029</b> |
| FAS <sup>b</sup><br>(median)                                                                            | 241       | 8.38       | 257     | 10.05      | 1.67                                | 17%                              | -            |
| <b>Rhinoconjunctivitis quality of life questionnaire (RQLQ(S)) score</b>                                |           |            |         |            |                                     |                                  |              |
| FAS <sup>b</sup><br>(adjusted mean)                                                                     | 229       | 1.38       | 240     | 1.58       | 0.19 <sup>e</sup><br>[0.02;0.37]    | 12%                              | <b>0.031</b> |
| FAS <sup>b</sup><br>(median)                                                                            | 229       | 1.25       | 240     | 1.46       | 0.21                                | 14%                              | -            |
| Post-hoc endpoints                                                                                      | N         | Proportion | N       | Proportion | Odds ratio <sup>f</sup><br>[95% CL] | p-value                          |              |
| <b>Probability of having a day with a rhinitis exacerbation</b>                                         |           |            |         |            |                                     |                                  |              |
| FAS (estimate) <sup>b</sup>                                                                             | 284       | 5.33%      | 298     | 11.14%     | 0.45<br>[0.28;0.72]                 | <b>0.001</b>                     |              |
| <b>Probability of having a day with a rhinitis exacerbation despite use of rhinitis pharmacotherapy</b> |           |            |         |            |                                     |                                  |              |
| FAS (estimate) <sup>b</sup>                                                                             | 284       | 3.43%      | 298     | 6.50%      | 0.51<br>[0.32;0.81]                 | <b>0.005</b>                     |              |

N: number of subjects in treatment group with data available for the analysis. CL: confidence limits

<sup>a</sup> FAS-MI: full analysis set with multiple imputations. The analysis treats subjects who discontinued the trial before the efficacy assessment period as placebo subjects. For the primary analysis (FAS-MI) only the absolute difference was pre-specified.

<sup>b</sup> FAS: full analysis set. All available data used to its full extent, i.e. subjects who provided data during the efficacy assessment period.

<sup>c</sup> Absolute difference: placebo minus 12 SQ-HDM, 95% confidence limits.

<sup>d</sup> Relative difference to placebo: placebo minus 12 SQ-HDM divided by placebo.

<sup>e</sup> The difference between 12 SQ-HDM and placebo was primarily driven by differences in three domains: sleep problems, practical problems and nose symptoms.

<sup>f</sup> Odds ratio for having a rhinitis exacerbation: 12 SQ-HDM over placebo.

### Supportive evidence – allergic rhinitis

A randomised, double-blind, placebo-controlled phase II trial was conducted in an allergen exposure chamber in 124 adults with house dust mite allergic rhinitis. Before each allergen challenge, subjects were washed out of all allergy pharmacotherapy. At the end-of-trial allergen challenge after 24 weeks of treatment with 12 SQ-HDM, 6 SQ-HDM or placebo, the mean rhinitis symptoms score was 7.45 [6.57;8.33] in the placebo group and 3.83 [2.94;4.72] in the 12 SQ-HDM group, corresponding to an absolute difference of 3.62 and a relative difference of 49% (95% confidence interval [35%;60%],  $p < 0.001$ ). The difference between 12 SQ-HDM and placebo was also statistically significant at 16 weeks (mean scores of 5.82 and 6.90, difference of 2.08 corresponding to 30%, 95% CI [17%;42%],  $p < 0.001$ ) and at 8 weeks (mean scores of 5.34 and 6.71, difference of 1.37 corresponding to 20%, 95% CI [7%;33%],  $p = 0.007$ ).

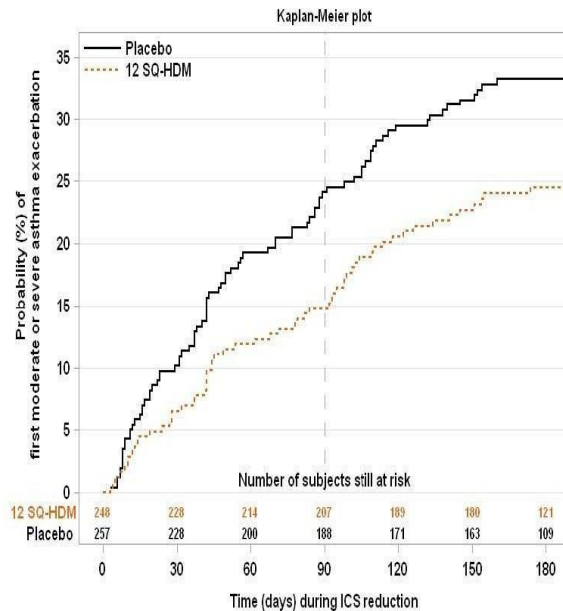
### **Allergic asthma**

#### The MITRA trial (MT-04)

The MITRA trial included 834 adults with house dust mite allergic asthma not well-controlled by daily use of inhaled corticosteroid (ICS) corresponding to 400-1200 µg budesonide. All subjects received 7-12 months' treatment with ACARIZAX® in addition to ICS and short-acting beta-agonist prior to ICS reduction. No titration phase to establish the lowest maintenance dose of ICS was conducted prior to randomisation. Efficacy was assessed by time to first moderate or severe asthma exacerbation under ICS reduction over the last 6 months of 13-18 months of treatment.

- The definition of a moderate asthma exacerbation was fulfilled if the subject experienced one or more of the 4 criteria below, and it led to change in treatment:
  - Nocturnal awakening or increase in symptoms: nocturnal awakening(s) due to asthma requiring short-acting  $\beta_2$  agonist (SABA) for two consecutive nights or increase of  $\geq 0.75$  from baseline in daily symptom score on two consecutive days.
  - Increased SABA use: increase from baseline in occasions of SABA use on two consecutive days (minimum increase: 4 puffs/day).
  - Deterioration in lung function:  $\geq 20\%$  decrease in PEF from baseline on at least two consecutive mornings/evenings or  $\geq 20\%$  decrease in FEV1 from baseline.

- Healthcare visit: visit to the emergency room / trial site for asthma treatment not requiring systemic corticosteroids.
- A severe asthma exacerbation was defined as experiencing at least one of the two following:
  - Need for systemic corticosteroids for  $\geq 3$  days
  - Emergency room visit requiring systemic corticosteroids or hospitalisation for  $\geq 12$ h.



The MITRA trial – illustration of primary efficacy data: Development over time of the risk for experiencing a moderate or severe asthma exacerbation during ICS reduction/withdrawal.

On the graph time = 0 represents the time of the ICS reduction to 50%. After approximately 3 months, i.e. at time = 90 days, ICS was completely withdrawn for those subjects who had not exacerbated.

| MITRA results                                               | 12 SQ-HDM |          | Placebo |          | Efficacy<br>12 SQ-HDM over placebo |                                | p-value |
|-------------------------------------------------------------|-----------|----------|---------|----------|------------------------------------|--------------------------------|---------|
|                                                             | N         | n (%)    | N       | n (%)    | Hazard<br>ratio<br>[95% CL]        | Risk<br>reduction <sup>a</sup> |         |
| Primary endpoint                                            |           |          |         |          |                                    |                                |         |
| Any exacerbation, mode-rate or severe (FAS-MI) <sup>b</sup> | 282       | 59 (21%) | 277     | 83 (30%) | 0.69<br>[0.50;0.96]                | 31%                            | 0.027   |
| Any exacerbation, mode-rate or severe (FAS) <sup>c</sup>    | 248       | 59 (24%) | 257     | 83 (32%) | 0.66<br>[0.47;0.93]                | 34%                            | 0.017   |
| Pre-defined analyses of components of the primary endpoint  |           |          |         |          |                                    |                                |         |
| Nocturnal awakening or increase in symptoms <sup>c</sup>    | 248       | 39 (16%) | 257     | 57 (22%) | 0.64<br>[0.42;0.96]                | 36%                            | 0.031   |
| Increased SABA use <sup>c</sup>                             | 248       | 18 (7%)  | 257     | 32 (12%) | 0.52<br>[0.29;0.94]                | 48%                            | 0.029   |
| Deterioration in lung function <sup>c</sup>                 | 248       | 30 (12%) | 257     | 45 (18%) | 0.58<br>[0.36;0.93]                | 42%                            | 0.022   |
| Severe exacerbation <sup>c</sup>                            | 248       | 10 (4%)  | 257     | 18 (7%)  | 0.49<br>[0.23;1.08]                | 51%                            | 0.076   |

N: number of subjects in treatment group with data available for the analysis.

n (%): number and percentage of subjects in treatment group meeting criterion.

CL: confidence limits

<sup>a</sup> Estimated by hazard ratio

<sup>b</sup> FAS-MI: full analysis set with multiple imputations. The analysis treats subjects who discontinued the trial before the efficacy assessment period as placebo subjects.

<sup>c</sup> FAS: full analysis set. All available data used to its full extent, i.e. including all subjects who provided data during the efficacy assessment period.

Post-hoc analyses of the asthma symptoms and symptomatic medication use in the last 4 weeks of the treatment period prior to reduction of inhaled corticosteroids were also conducted to investigate the effect of ACARIZAX<sup>®</sup> as add-on to inhaled corticosteroid. The analyses looked at asthma daytime and nocturnal symptom scores, nocturnal awakenings, and SABA intake. The post-hoc analyses showed numerical differences consistently in favour of 12 SQ-HDM over placebo for all parameters investigated during the last 4 weeks prior to inhaled corticosteroid reduction. The differences were only statistically significant for the asthma daytime symptom score ( $p=0.0450$ ) and the odds for no nocturnal awakenings ( $p=0.0409$ ).

#### Supportive evidence – allergic asthma

In a double-blind, randomised, placebo-controlled phase II trial, 604 subjects  $\geq 14$  years old with house dust mite allergic asthma controlled by inhaled corticosteroids (100-800 $\mu$ g budesonide) and a clinical history of house dust mite allergic rhinitis were randomised to approximately 1 year of treatment with 1, 3 or 6 SQ-HDM or placebo. At the 4-week end-of-trial efficacy evaluation period, the mean change from baseline in the daily ICS dose was 207.6  $\mu$ g budesonide in the 6 SQ-HDM group and 126.3  $\mu$ g in the placebo group corresponding to an absolute difference of 81  $\mu$ g budesonide per day (95% confidence interval [27;136],  $p=0.004$ ). Relative mean and median ICS reductions from baseline were 42% and 50% for 6 SQ-HDM and 15% and 25% for placebo. In a post-hoc analysis of a subgroup ( $N=108$ ) of subjects with lower asthma control and ICS  $\geq 400$   $\mu$ g budesonide, the mean change from baseline in the daily ICS dose was 384.4  $\mu$ g budesonide in the 6 SQ-HDM group and 57.8  $\mu$ g in the placebo group corresponding to an absolute difference between 6 SQ-HDM and placebo of 327  $\mu$ g budesonide per day (95% CI [182;471],  $p<0.0001$ , post-hoc analysis).

### **Paediatric population**

#### Children 5-11 years of age

ACARIZAX<sup>®</sup> is not indicated in children below 12 years of age (see section 4.2). Limited safety and tolerability data exist for children 5-11 years of age.

#### Adolescents 12-17 years of age

ACARIZAX<sup>®</sup> is indicated for treatment of allergic rhinitis in adolescents. Efficacy and safety in adolescents and adults have been shown to be similar.

ACARIZAX<sup>®</sup> is not indicated for treatment of allergic asthma in adolescents.

#### Elderly population

ACARIZAX<sup>®</sup> is not indicated in patients  $>65$  years of age (see section 4.2). Limited safety and tolerability data exist for elderly patients  $>65$  years of age.

#### Long-term treatment

International treatment guidelines refer to a treatment period of 3 years for allergy immunotherapy to achieve disease modification. Efficacy data is available for 18 months of treatment with ACARIZAX<sup>®</sup> from the MITRA trial. Long-term efficacy has not been established.

### **5.2 Pharmacokinetic properties**

No clinical studies investigating the pharmacokinetic profile and metabolism of ACARIZAX<sup>®</sup> have been conducted. The effect of allergy immunotherapy is mediated through immunological mechanisms, and there is limited information available on the pharmacokinetic properties.

The active molecules of an allergen extract are composed primarily of proteins. For sublingually administered allergy immunotherapy products, studies have shown that no passive absorption of the allergen through the oral mucosa occurs. Evidence points towards the allergen being taken up through the

oral mucosa by dendritic cells, in particular Langerhans cells. Allergen which is not absorbed in this manner is expected to be hydrolysed to amino acids and small polypeptides in the lumen of the gastrointestinal tract. There is no evidence to suggest that the allergens present in ACARIZAX<sup>®</sup> are absorbed into the vascular system after sublingual administration to any significant extent.

#### **Preclinical safety data**

Conventional studies of general toxicology and toxicity to reproduction in mice have revealed no special hazards to humans.

### **6 PHARMACEUTICAL PARTICULARS**

#### **6.1 List of excipients**

Gelatine (fish source)

Mannitol

Sodium hydroxide (for pH adjustment)

#### **6.2 Incompatibilities**

Not applicable

#### **6.3 Shelf life**

Please refer to the expiry printed on the carton.

#### **6.4 Special precautions for storage**

Store below 30°C

#### **6.5 Nature and contents of container**

Aluminium/aluminium blister cards in outer carton. Each blister card contains 10 sublingual lyophilisates. Pack sizes: 10 and 30.

#### **6.6 Special precautions for disposal**

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

### **7 PRODUCT REGISTRATION HOLDER & IMPORTER**

Abbott Laboratories (M) Sdn. Bhd.

27-02, Level 27, Imazium,

No. 8, Jalan SS 21/37, Damansara Uptown,

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